MOOSE Checklist for Meta-analyses of Observational Studies

Item No	Recommendation	Reported on Page No
Reporting of background should include		
1	Problem definition	p. 1 – Introduction
2	Hypothesis statement	p. 1 – Introduction
3	Description of study outcome(s)	Table I
4	Type of exposure or intervention used	Table I
5	Type of study designs used	Table I
6	Study population	Table I
Reporting of search strategy should include		
7	Qualifications of searchers (eg, librarians and investigators)	p. 3 – Methods
8	Search strategy, including time period included in the synthesis and key words	p. 3 – Methods/search strategy and data sources
9	Effort to include all available studies, including contact with authors	p. 3 – Methods/study selection
10	Databases and registries searched	p. 3 – Methods/search strategy and data sources
11	Search software used, name and version, including special features used (eg, explosion)	p. 3 – Methods/search strategy and data sources + Study protocol[1] published
12	Use of hand searching (eg, reference lists of obtained articles)	p. 5 – Methods/study selection
13	List of citations located and those excluded, including justification	Figure 1: PRISMA flow diagram
14	Method of addressing articles published in languages other than English	German included p. 3 – Methods/search strategy and data sources
15	Method of handling abstracts and unpublished studies	n.a.
16	Description of any contact with authors	p. 4 – Methods/study selection
Reporting of methods should include		
17	Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested	Criteria defined in published study protocol[1]

[1] Find online: https://www.crd.york.ac.uk/prospero/ Protocol ID: CRD42020165356

18	Rationale for the selection and coding of data (eg, sound clinical principles or convenience)	Criteria defined in published study protocol[1]
19	Documentation of how data were classified and coded (eg, multiple raters, blinding and interrater reliability)	p. 4 – Methods/study selection
20	Assessment of confounding (eg, comparability of cases and controls in studies where appropriate)	n.a.
21	Assessment of study quality, including blinding of quality assessors, stratification or regression on possible predictors of study results	p. 4 – Methods/Quality assessment
22	Assessment of heterogeneity	n.a. since narrative
23	Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated	p.4 – Methods/ Data extraction methods and data synthesis + Quality assessment
24	Provision of appropriate tables and graphics	n=5 tables n=1 graphic
Reporting of results should include		
25	Graphic summarizing individual study estimates and overall estimate	Table II Estimate n.a.
26	Table giving descriptive information for each study included	Table II
27	Results of sensitivity testing (eg, subgroup analysis)	Subgroup analysis Table IV Sensitivity testing n.a. since narrative
28	Indication of statistical uncertainty of findings	n.a. since narrative

Item No	Recommendation	Reported on Page No
Reporting of discussion should include		
29	Quantitative assessment of bias (eg, publication bias)	n.a. since narrative, no effects tested
30	Justification for exclusion (eg, exclusion of non-English language citations)	n.a.
31	Assessment of quality of included studies	Table V
Reporting of conclusions should include		
32	Consideration of alternative explanations for observed results	p. 9 – Discussion
33	Generalization of the conclusions (ie, appropriate for the data presented and within the domain of the literature review)	n.a. since narrative
34	Guidelines for future research	pp. 13-14 checklist as supplemental material
35	Disclosure of funding source	Title page

From: Stroup DF, Berlin JA, Morton SC, et al, for the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) Group. Meta-analysis of Observational Studies in Epidemiology. A Proposal for Reporting. *JAMA*. 2000;283(15):2008-2012. doi: 10.1001/jama.283.15.2008.